In an age of AI, we still need people.

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How do I view AI?

AI is transforming how we work

AI is like an infinite number of monkeys sitting in front of an infinite number of laptops (equipped with a Python or R IDE...)



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AI is transforming how we work

It's a 10x transformer for the ***speed*** at which we write code.

But also the ***amount***.

Those monkeys are ***BUSY***.



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AI is transforming how we work

I'm not saying the code is ***bad***, but I ***am*** saying that Jenny Bryan might have opinions about the code smells and feels...



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What are my experiences (so far) with using AI for data science?

My personal experiences (1)

I’m astounded at AI assisted exploratory data analysis...

Useful insights, traceability.

Fast.

`/report` is 👍



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My personal experiences (2)

Continue using your good practice workflows:

1. Write code
2. Turn it into a function
3. Test and document that function
4. Use that function in a Shiny app.



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My personal experiences (2)

Shiny apps can become single-use and “disposable”

...rather than “build it once, build it well, deploy it, let others use it.”



My personal experiences (3)

Refactoring code:

AI did a LOT of work quickly over several iterations. Generated a TON of code and (wordy!) documentation...



My personal experiences (3)

Refactoring code:

...I'm still working through what it has done, trying to understand and evaluate it.



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My personal experiences (3)

Refactoring code:

STOP. THINK.
DEFINE / REFINE THE DESIGN PRINCIPLES.

Get AI to ask you questions based on code and documentation... Gain better understanding.



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My personal experiences (4)

Skill definitions are going to be transformative.

Can we capture and codify our workflows and knowledge so that agents produce content more aligned with “how we work”?

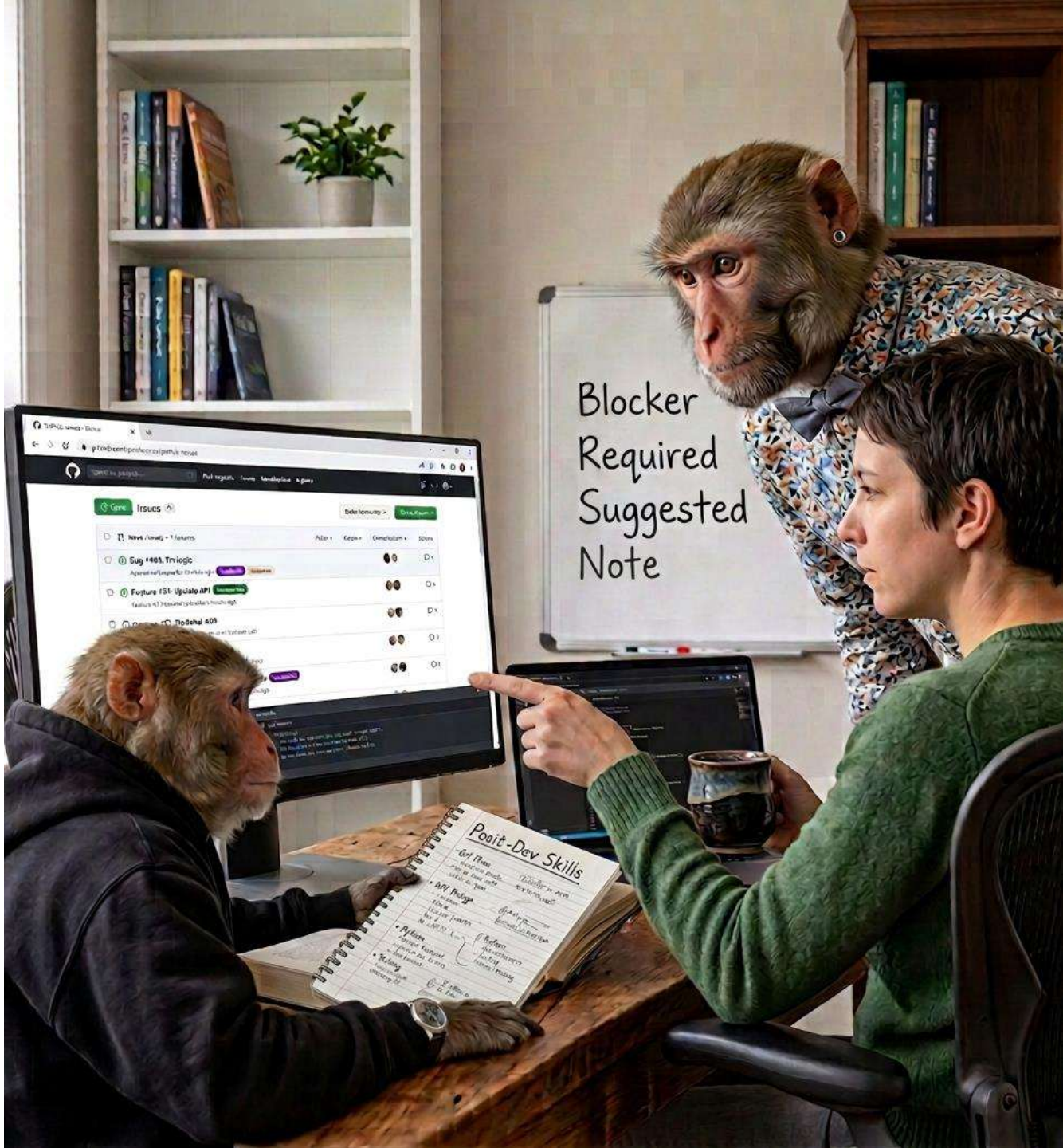


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My personal experiences (4)

Using Posit-Dev skill “critical-code-reviewer” is like having Posit developers helping with your code review.

BTW: Kinda chic to use {gh} package to create GitHub Issues off the back of the findings…



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Questions and **concerns I have...**

FAST to PoC

We can go ***REALLY*** quickly from nothing to something.

Potential for Agile development is huge.

Customer can get hands on with a functioning prototype and provide feedback ***QUICKLY***.



BUT THEN...

Going from PoC to Production code: we may need to refactor quite a lot of content to make it more sustainable and manageable.



Functional but not maintainable (by humans)

DOES IT MATTER how it's built as long as the end product ***WORKS***, even if we don't know ***HOW***?

Presumably it can be fixed later (by AI) if it breaks...



Vibe coding

Confidently wrong.

No idea ***why*** it works.

No idea ***why*** it went wrong.

Confidently fixing things at random.

Colleagues should have a “license to use AI” – minimal R or Python skillset before use.



AI bots talking to AI bots

If we leave it to an AI agent to review code that another AI has written then are we really winning?

I worry about us devolving human review and ***oversight*** to AI...



Organisations devoid of developers

Some companies are going to decide that they don't need "ground level" developers any more.

Experienced developers become AI "code checkers", architects, etc.

But then, how do you grow the next generation of experienced developers?



Where people still **matter...**

Where people still matter

Learning together with someone else makes the journey easier...

Building and engaging in communities shares knowledge and skills.

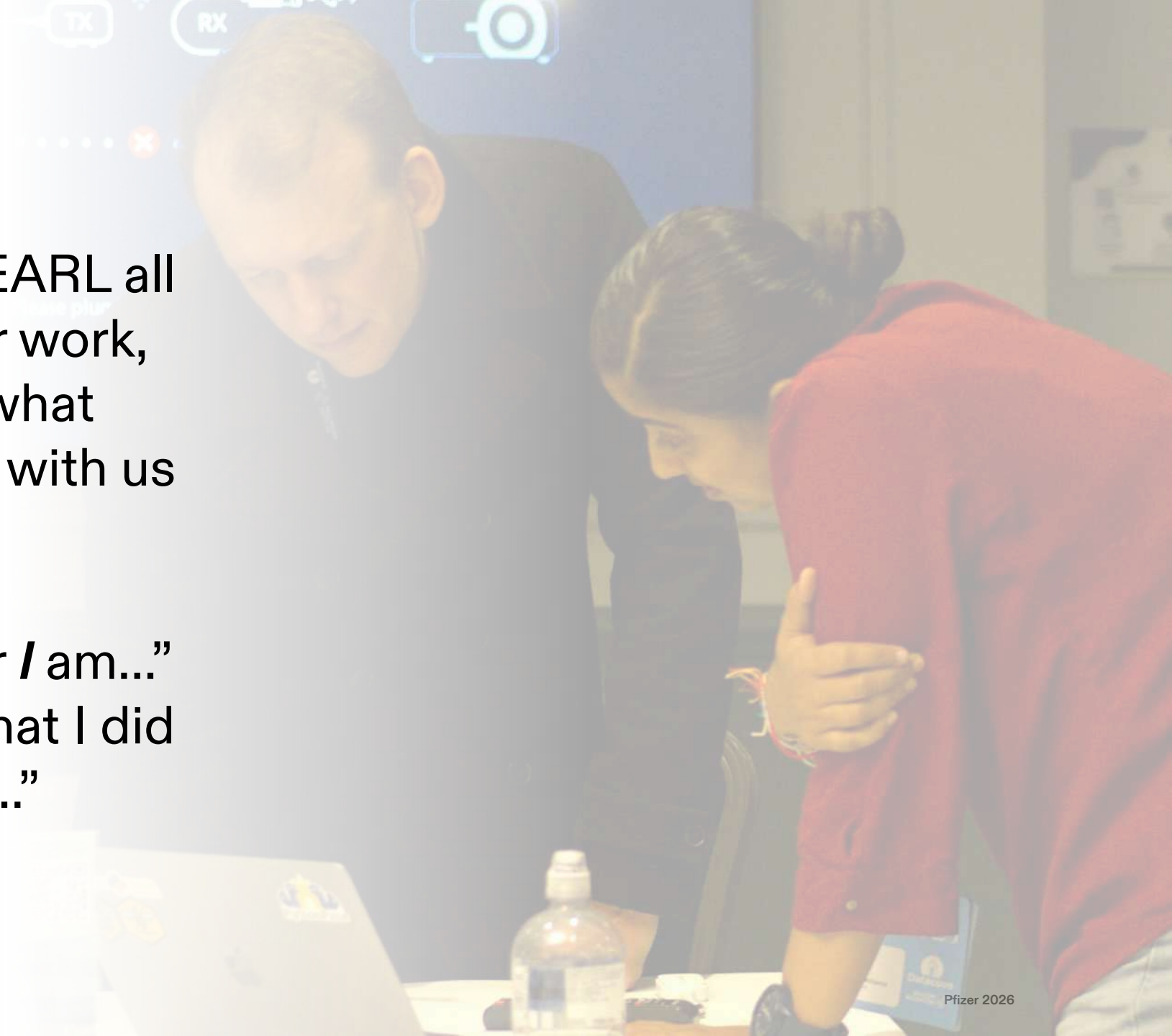
Open Source by definition means that we like to share what we've made.



Where people still matter

The presenters here at EARL all have thought about their work, and are happy to share what worked, and what didn't with us all.

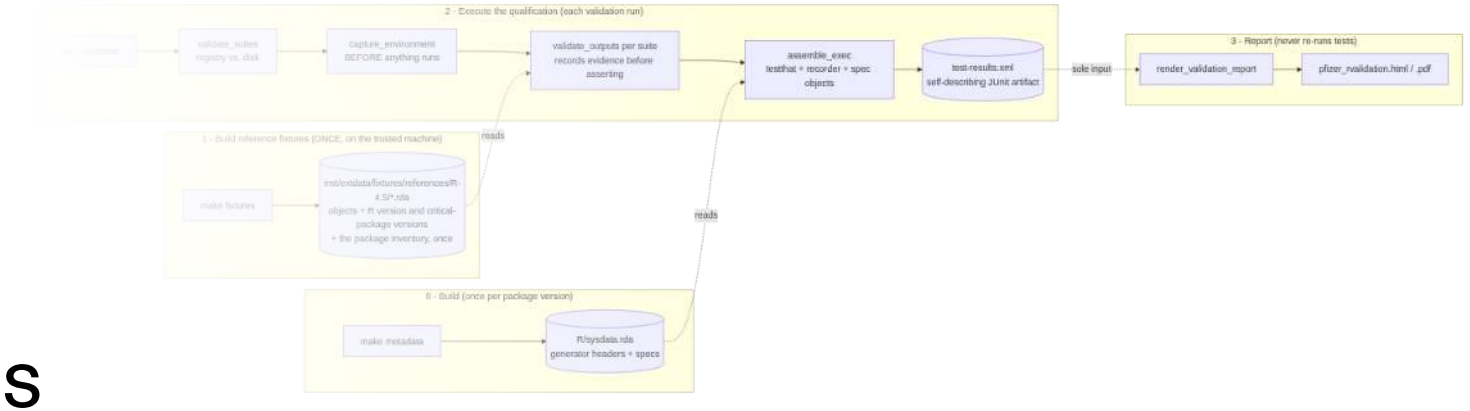
It's not "Look how clever ***I*** am..."
It's "Let me show you what I did so you can do the same..."



Where people still matter

People are good at refactoring code and making “architectural” decisions.

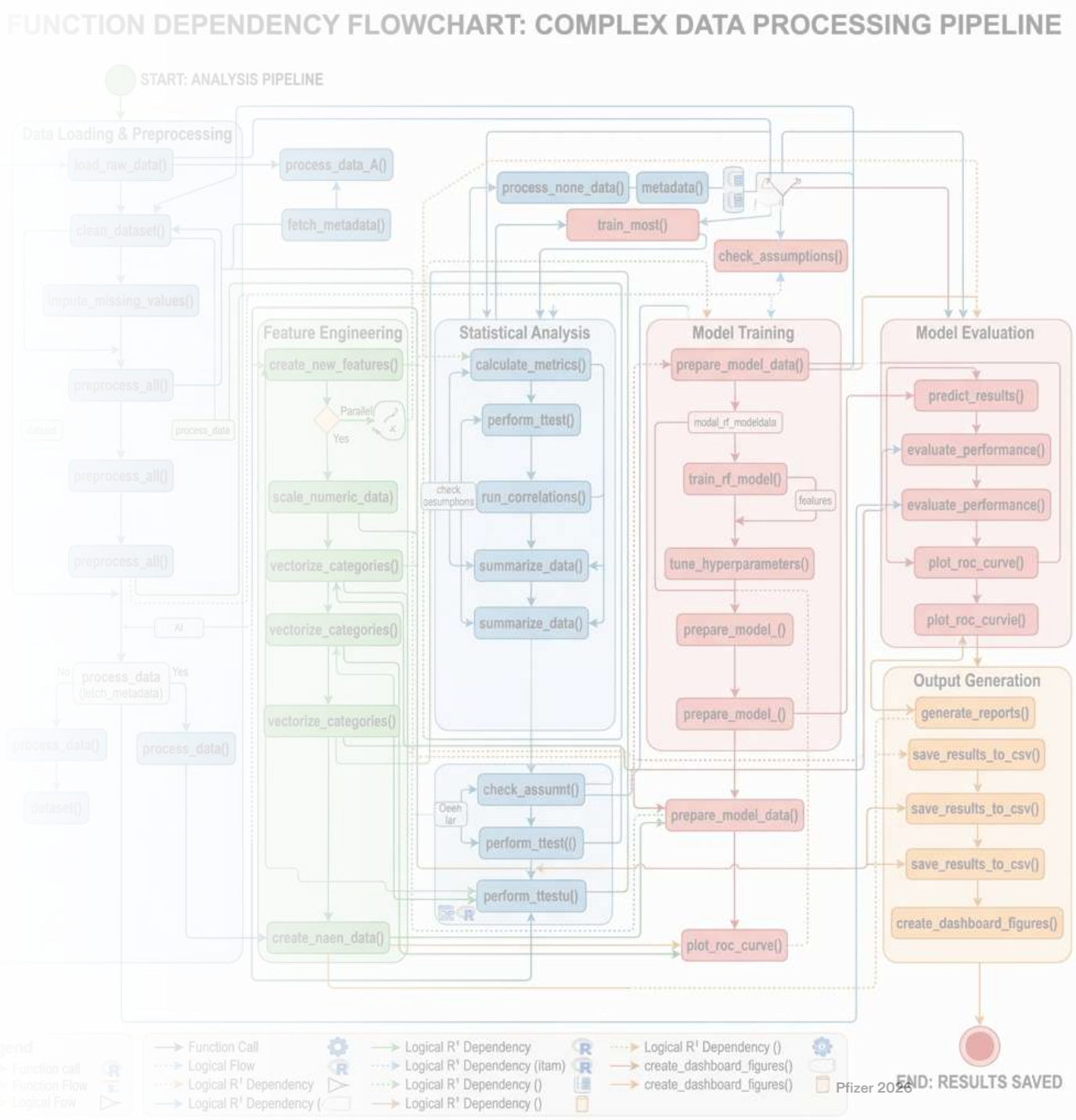
Again, experience tells us what is likely to work, technical constraints, future proofing.



Where people still matter

Experienced coders know what code is brittle, will be difficult to maintain, introduces technical debt.

Today's technical debt is tomorrow's nightmare during crunch timelines.

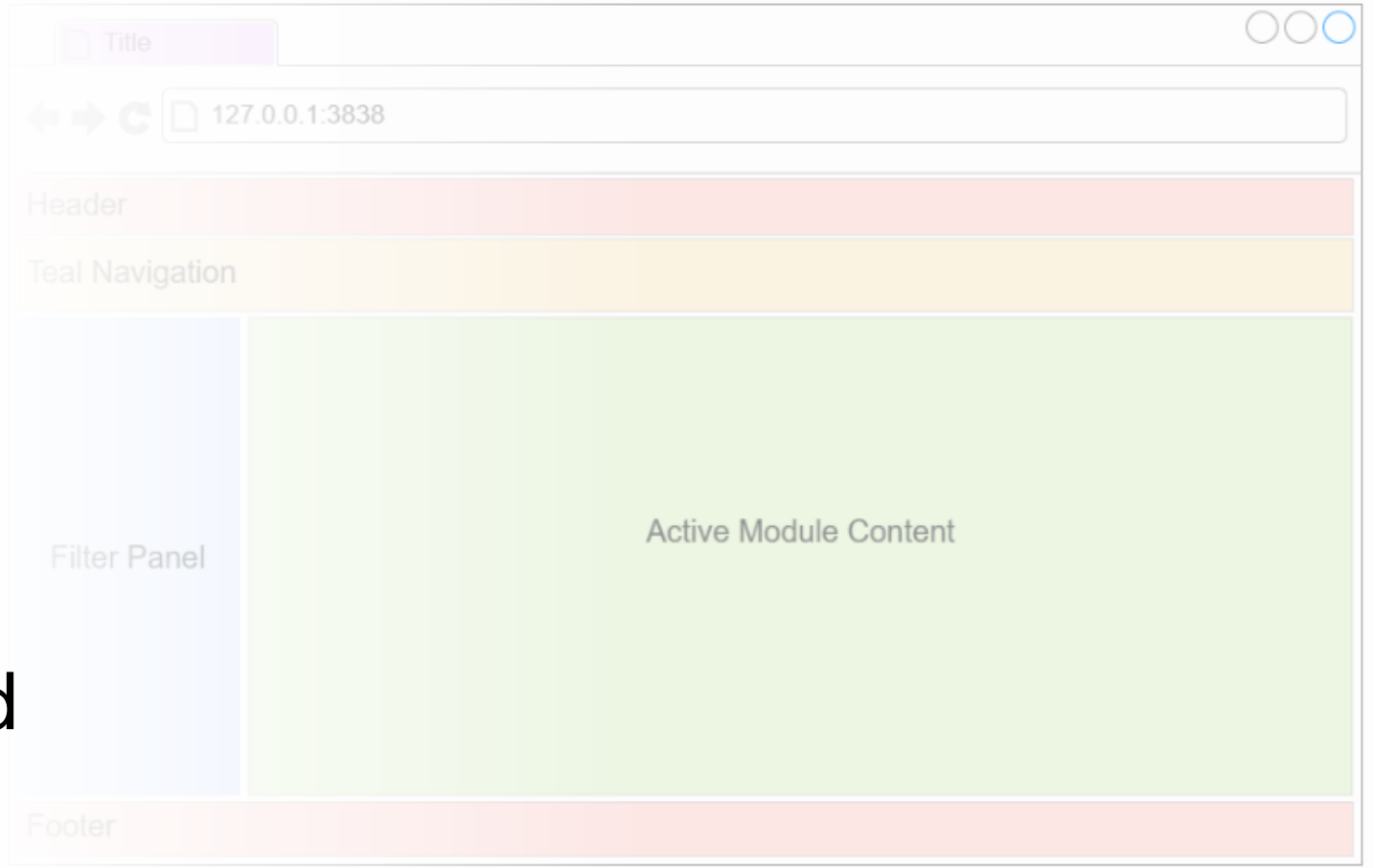


Where people still matter

People know UI / UX.

How can I make this easier? How do I make it obvious what to do next?

Also, “Wow this is ugly and complicated... we need to do better.”

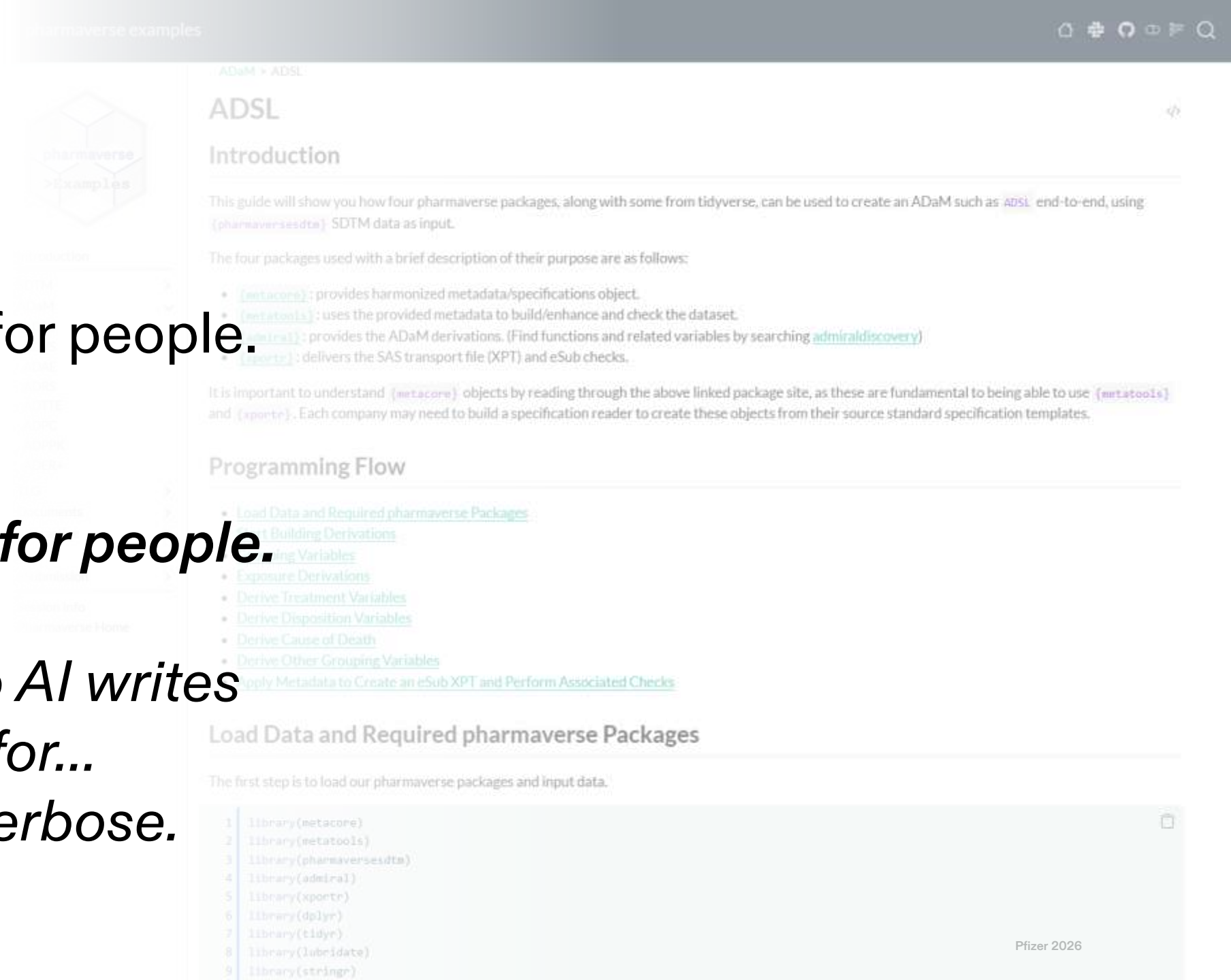


Where people still matter

People write documentation for people.

Please write documentation for people.

I'm not sure who AI writes documentation for…
Unnecessarily verbose.



Where people still matter

People tend to have REALLY GOOD curated lists of blog posts, podcasts, writers, people doing good and creative work, inspiring content.

Conversations over coffee at conference are often the best bits...



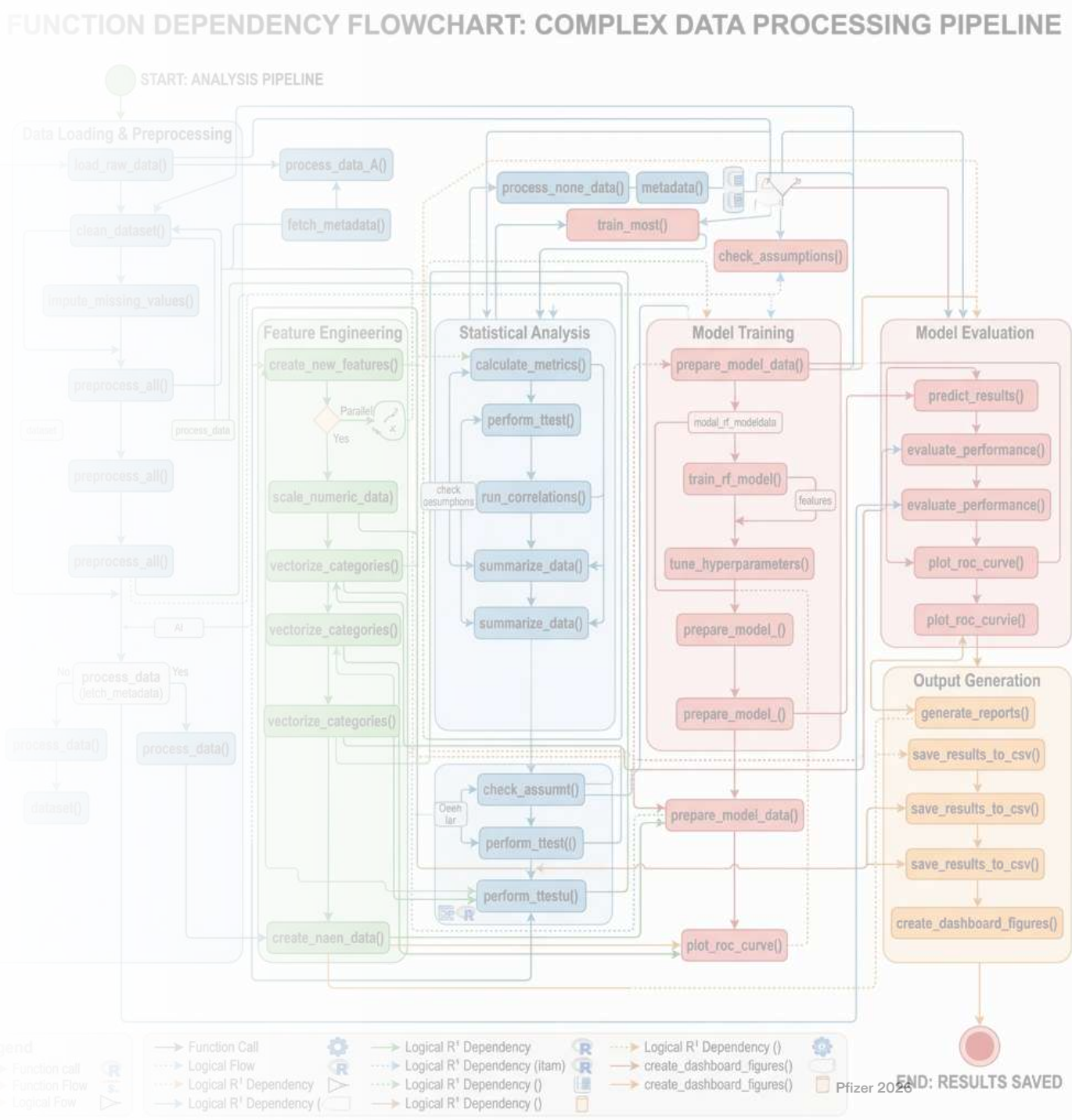
A call to action...

What can I do?

AI models are built off open source repositories and contributions.

We need to keep the quality of that code base ***HIGH***. AI slop in code and documentation “poisons the well”.

Contribute. Improve. Share.

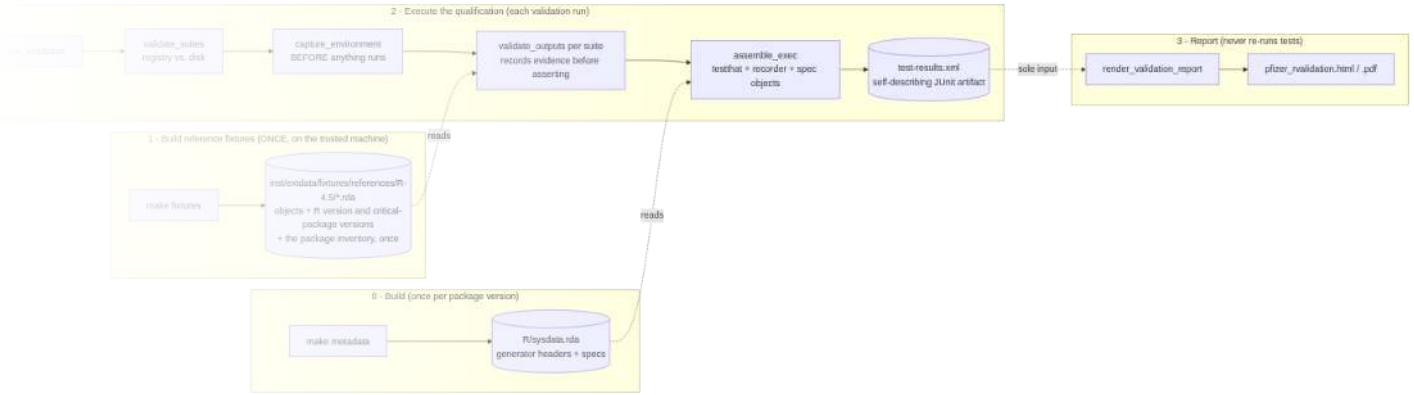


What can I do?

Get better at making good choices… Stamp out technical debt.

Don’t accept “quick and dirty”. There’s no need anymore.

Define “What A Good One Looks Like” Skill for the AI…



What can I do?

Be the person who helps someone else to learn.

Share what you know & share your experience.

YOU DO NOT NEED TO BE "EXPERT" LEVEL TO DO THIS!
"LEARN OUT LOUD"

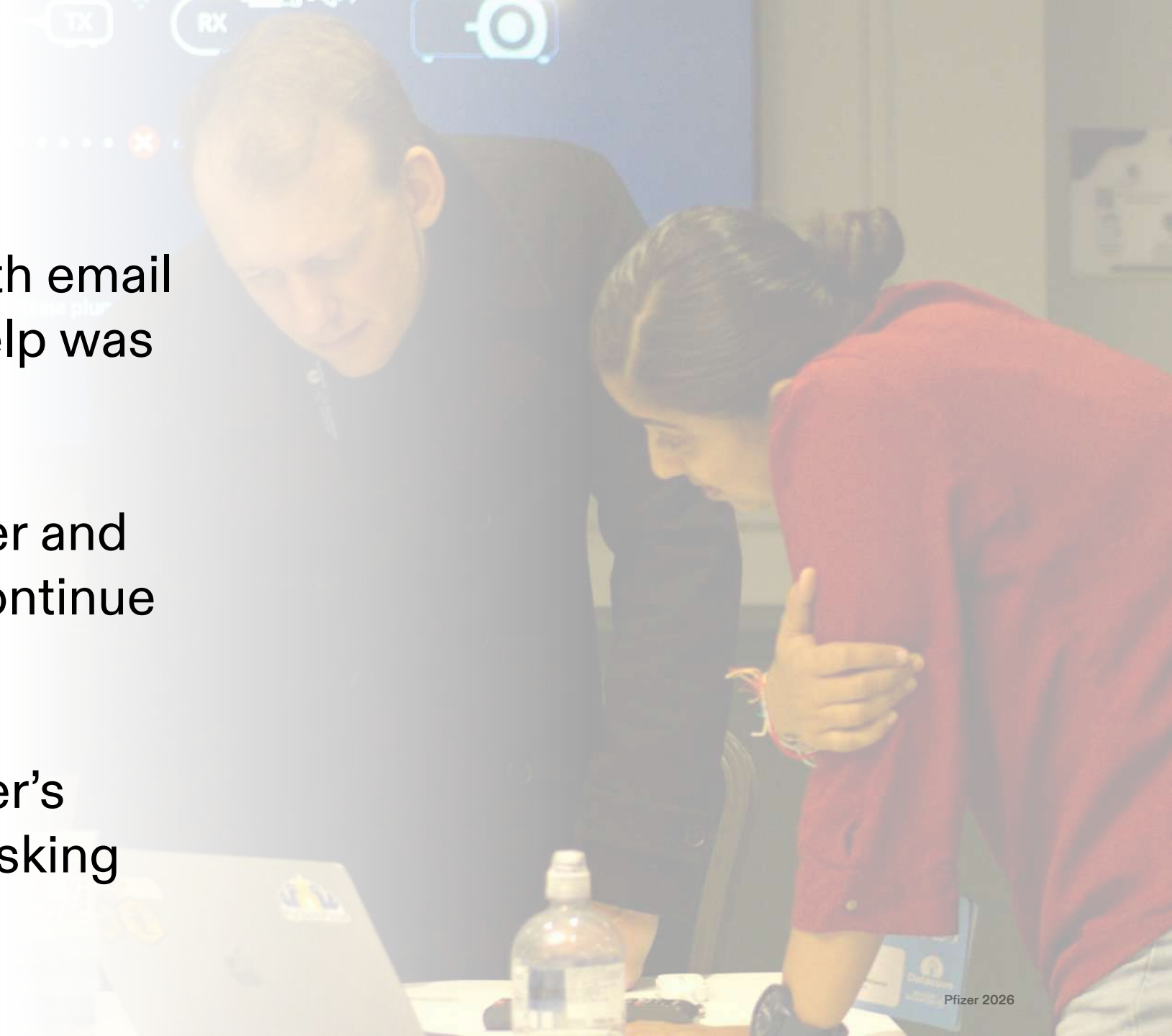


What can I do?

Our early experience with email distribution lists for R help was pretty unforgiving...

We've grown to do better and be more helpful. Let's continue that...

Put yourself in a beginner's shoes... ***WHY*** are they asking that question?



AI good practices

Go slower

Spend more time in planning and thinking

Capture that design process (as markdown)

Don’t rush to write code



Personal development in the age of AI

Remind yourself periodically how to do it “by hand”.

Cat Hicks has written skills to help you ***learn as you use AI***

Keep up to date with R / Python e.g. R Weekly

Learn more about AI prompting, Skills, Agentic approaches.



Where people still matter

Be part of the ***community***.

Help grow that ***community.***

Be part of the future of R…
I'd like it to still be relevant in another 30 years….



THANK YOU!

And thanks to Datacove
for the invitation...